Sharing and Publishing

Workflows for dynamically generated reports

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Sharing scenarios

These are some frequent use cases:

1. `Preliminary results` (temporary, internal reports)
2. `Internal monitoring` (regularly updated reports)
3. `Templates or documentation`
4. `Dissemination` to broader audiences
5. `Publication` (preprint, dataset, article, etc)

Git version-controlled repositories

- Git: free and open source distributed version control system
- Popular platforms: GitLab (open) and GitHub (Microsoft)
- There are `GitLab instances` at many universities
- Meant for code and documents, not as data repositories
- Recommended for `all` sharing scenarios

Sharing with GitLab pages

- Quarto websites can easily assemble HTMLs in a website
- GitLab pages can host multiple `static` websites for free
- They can be `private` or `public`
- HTML interactivity: e.g., `dashboards`, `plotly`, `DataTables`
- Example use-cases: sharing results, technical documentation, templates, dissemination, etc.

Persistent identification

A persistent identifier (e.g., DOI) enable our reports to be `reused` and `found` in time.

Required to make our resources `citable`

Where can I get a DOI ?

- Data repositories (e.g., Zenodo, OSF, Ebrains…)
- Journals and preprint platforms

Popular generalist DOI-enabling repositories

Zenodo

- Free. 50 GB upload limit per dataset (up to 200Gb upon request).
- Hosted by CERN. The data stays in Europe.

Open Science Framework (OSF)

- Free. Private projects limited to 5 GB and public projects to 50 GB.
- Default storage location in US, but also servers in Canada, Germany and Australia.

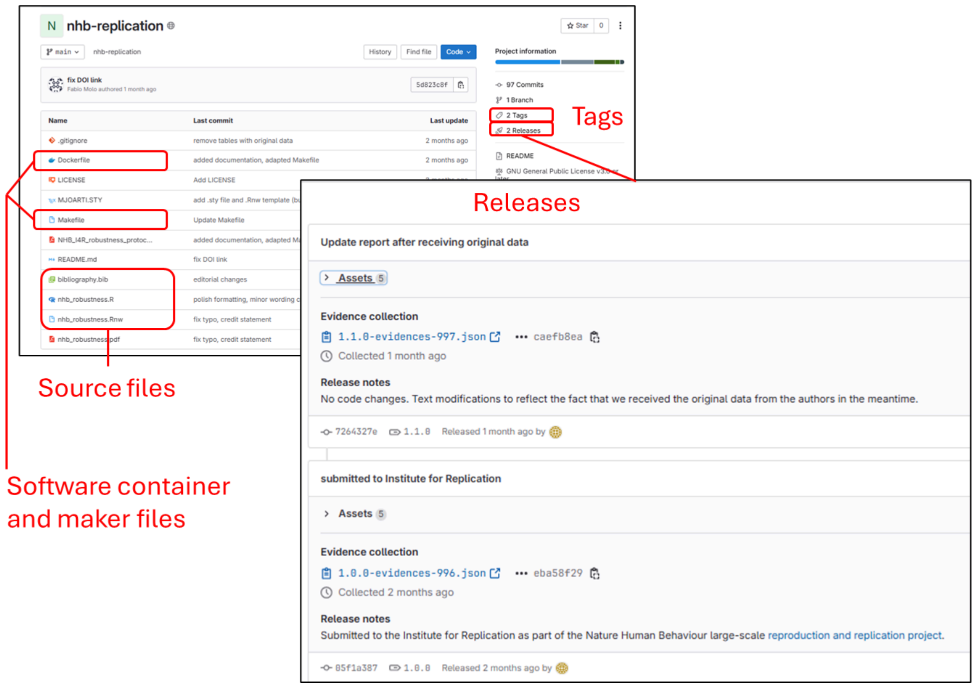
Git repositories and DOIs

- GitLab and GitHub do not offer DOIs
- ‘Persistent URLs’/‘Permanent links’ not as strongly persistent
- But we can combine a GitLab releases with DOI enabling platforms
- A `release` in GitLab is a `snapshot` of a repository with
  - Version tag
  - Source code and metadata

Workflow example with GitLab and Zenodo

Robustness check for Nature Human Behaviour

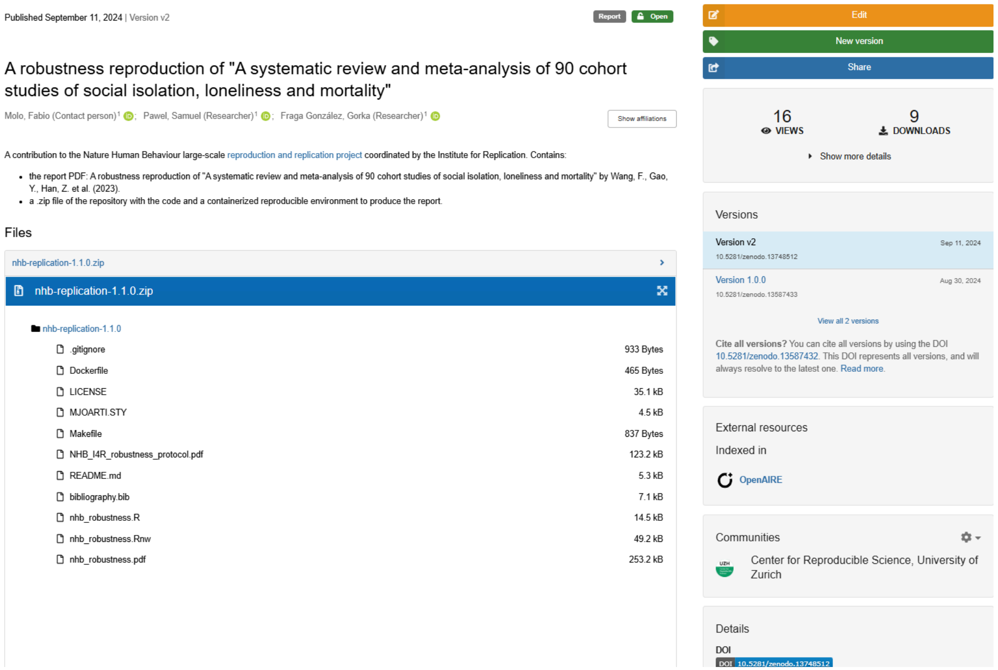
GitLab Repository



Each release includes assets (zipped source code) and a .json with the release metadata

The README.md file has a link to the Zenodo entry with a DOI and citation information.

Zenodo entry



A citable Zenodo entry with a DOI and the content of the repository release in a .zip file

Preregistration in Open Science Framework (OSF)

- OSF feature to preregister your research plan
- It can be private and released upon project completion
- OSF templates can help structure your report
- Enhances visibility, rigor and transparency of your work

Publication workflow

Git repository (e.g., Gitlab)

- Active code development
- Version tagged 'releases'
- Link to Zenodo publication

Archive with DOI (e.g., Zenodo entry)

Files uploaded

- Release (e.g., .zip) with version tag
- New uploads assign a new DOI
- But there is a DOI to find all versions

Metadata

- **URL** to Git repository ('software' field)
- It can be edited without a new DOI

Journal publications

Citing the DOI entry with code

ThanK-YoU